

LE MANS ARNAGE AP, France

WMO: 072350

Lat: 47.9458N

Lon: 0.1942E

Elev: 59

StdP: 100.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	-2.9	-8.2	1.9	-1.9	-6.1	2.3	-1.2	9.2	10.1	8.3	9.7	2.0	30	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.9	20.2	29.7	19.4	27.7	18.7	21.5	29.0	20.5	27.4	19.6	25.8	3.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	13.9	23.7	18.1	13.1	23.1	17.2	12.4	22.0	62.4	29.2	59.0	27.4	55.9	25.9	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.8	6.1	-7.7	35.1	2.9	1.9	-9.8	36.5	-11.5	37.7	-13.1	38.7	-15.1	40.2		
(5)			-8.1	23.3	2.9	1.2	-10.2	24.2	-11.9	24.9	-13.5	25.5	-15.5	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	5.2	5.9	8.4	11.0	14.6	18.1	20.1	19.8	16.5	13.0	8.4	5.7		
	DBStd	6.33	4.00	3.71	3.24	3.26	3.29	3.43	3.14	3.13	2.91	3.36	3.47	3.98		
	HDD10.0	588	151	119	69	25	3	0	0	0	0	12	69	139		
	HDD18.3	2413	406	348	307	220	124	46	17	18	71	168	299	391		
	CDD10.0	1413	4	4	21	56	144	242	313	304	195	104	21	7		
	CDD18.3	197	0	0	0	1	7	37	72	63	15	2	0	0		
	CDH23.3	1943	0	0	0	19	89	383	683	609	149	11	0	0		
	CDH26.7	639	0	0	0	1	14	123	238	227	35	1	0	0		
(14) Wind	WSAvg	2.9	3.0	3.1	3.1	3.1	3.0	2.9	2.8	2.5	2.6	2.5	2.7	3.0		
(15) Precipitation	PrecAvg	682	65	58	60	50	59	46	50	45	55	60	70	64		
	PrecMax	957	128	115	102	139	137	108	121	146	118	210	153	136		
	PrecMin	464	20	7	3	3	12	0	13	6	10	5	13	10		
	PrecStd	110	29	28	29	30	33	25	27	32	34	45	37	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.8	21.3	25.7	28.2	33.8	34.4	35.3	30.1	24.6	17.9	14.6		
		MCWB	12.3	11.3	12.8	15.7	18.4	21.2	21.0	21.7	18.9	17.7	14.6	12.6		
	2%	DB	13.1	13.9	18.2	22.7	25.9	30.0	31.6	31.9	27.2	21.9	15.9	13.5		
		MCWB	11.6	10.3	11.7	14.5	17.0	19.8	20.1	20.4	18.2	16.7	13.7	12.0		
	5%	DB	12.0	12.3	15.9	20.0	23.6	27.5	29.4	29.2	24.8	19.7	14.7	12.6		
		MCWB	10.5	9.9	11.2	13.1	16.1	18.9	19.2	19.6	17.3	15.7	12.8	11.2		
	10%	DB	10.7	11.2	14.1	17.4	21.4	25.2	27.3	26.7	22.7	18.1	13.5	11.5		
		MCWB	9.3	9.3	10.5	11.7	15.1	17.6	18.6	18.5	16.2	15.0	11.7	10.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.3	14.1	17.0	19.6	22.6	22.6	23.0	20.5	18.6	15.6	13.4		
		MCDB	13.6	14.7	18.8	24.0	26.2	30.5	30.8	32.4	26.7	22.4	16.9	14.1		
	2%	WB	11.9	11.2	12.8	15.1	18.2	20.7	21.1	21.6	19.2	17.4	14.1	12.3		
		MCDB	12.8	12.9	16.0	21.2	24.0	27.9	29.0	29.1	24.8	20.4	15.5	13.2		
	5%	WB	10.7	10.5	11.9	13.7	16.8	19.5	20.1	20.5	18.1	16.5	13.0	11.4		
		MCDB	11.8	12.0	14.8	18.7	22.0	25.9	27.3	27.1	23.5	18.9	14.4	12.4		
	10%	WB	9.5	9.5	11.0	12.4	15.6	18.4	19.2	19.4	17.1	15.6	12.0	10.4		
		MCDB	10.6	11.0	13.6	16.4	20.5	24.0	25.5	25.0	21.1	17.7	13.3	11.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	7.4	9.1	10.3	10.2	10.7	11.0	11.3	10.8	8.7	6.8	5.9		
		MCDBR	6.1	8.7	12.4	14.7	13.7	14.2	14.5	15.5	14.4	10.8	7.4	5.7		
	5% WB	MCWBR	5.0	6.1	7.2	7.4	6.5	6.0	5.3	5.9	6.6	6.1	5.3	4.7		
		MCWBR	5.9	7.6	9.8	12.8	12.3	12.6	12.8	12.9	12.1	9.1	6.5	5.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.325	0.343	0.373	0.397	0.401	0.409	0.403	0.392	0.368	0.362	0.334	0.319		
		taud	2.431	2.379	2.294	2.232	2.266	2.282	2.296	2.346	2.391	2.414	2.436	2.408		
	Ebn at Noon	Ebn at Noon	746	810	838	852	862	855	855	849	836	777	725	699		
		Edh at Noon	67	87	110	129	130	129	125	114	99	83	66	61		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.00	1.82	2.98	4.42	5.21	5.82	5.57	4.89	3.81	2.24	1.23	0.86		
	RadStd	RadStd	0.10	0.23	0.37	0.51	0.45	0.57	0.37	0.45	0.28	0.19	0.12	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	Regional (34 neighbors)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page